

BRIDGEPULSE · REQUIREMENTS

BridgePulse requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	R12 redefined per BRP-DDR-001 D1; status of every requirement from BRP-CAL-001; interface assumptions for FieldNode
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

BridgePulse requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with bridge owners or inspectors, and will be revised after co-design sessions (see BRP-PRB-001). Status is now judged by calculation in BRP-CAL-001 against the parametric model and the priced BOM; "met on paper" is not a verification by test.

Reference installation. A steel or concrete footbridge or short-span road bridge of 5 to 30 m span, monitored at midspan, in a climate from −20 to +50 °C, with LoRaWAN coverage and no mains power. The worked example is the 7 m steel footbridge in BRP-PRC-001.

Changes in v0.3. R12 is redefined to cost the BridgePulse-specific parts, with the FieldNode core costed in its own repo and the complete cost stated beside it (BRP-DDR-001, D1, adopted as recommended for TRL 3 under Amish's 2026-09-25 instruction, open for his review). No other target changes.

Table 1. Requirements and status at TRL 3 (BRP-CAL-001, Table 5).

ID	Requirement	Target	Verification	Status at TRL 3
R1	Measure bridge acceleration	3 axes; noise density 25 $\mu\text{g}/\text{VHz}$ or less; band 0.5 to 60 Hz	Datasheet; later bench noise test	Met by design (22.5 $\mu\text{g}/\text{VHz}$; 62.5 Hz filter corner)
R2	Track natural frequencies	Every mode with a clear peak below 60 Hz, at least the first vertical mode; bin width 0.02 Hz or less per hourly record; repeatability 0.2 % (1σ) or better over a day of steady temperature	Spectral simulation (BRP-CAL-001, B); later field data	Not met on the example footbridge: sensor-noise scatter 0.053 % at 100 μg with a curve fit, but walker mass adds 0.45 % at 5 crossings an hour
R3	Flag structural change	Flag a sustained shift of 1 % or more in a tracked frequency, after temperature compensation, within 14 days; one false flag per bridge per year or fewer	Detection statistics (BRP-CAL-001, C); then public benchmark data and a field baseline	Not verifiable at TRL 3: needs a daily residual of 0.75 % or less after temperature and load compensation
R4	Measure strain	2 channels; range $\pm 1,000 \mu\epsilon$; resolution 2 $\mu\epsilon$ or better; drift 5 $\mu\epsilon$ per month or less after temperature compensation	ADC noise calculation; later outdoor drift test	Not verifiable at TRL 3 for drift; resolution (0.09 $\mu\epsilon$ RMS) and range met on paper
R5	Measure temperature	Steel surface and shaded air; $\pm 0.5^\circ\text{C}$ or better, checked against a reference	Datasheet; CalRig chamber check	At risk: $\pm 0.5^\circ\text{C}$ only from -10°C upward
R6	Stay within the FieldNode energy budget	Average draw 30 mW or less at the FieldNode sensor port	Power calculation; later current logging	Met on paper (16.9 mW)
R7	Send and keep the data	One summary per hour of 48 bytes or less, within LoRaWAN regional limits at the data rate in use; raw records kept on site for 12 months or more	Payload and airtime calculation; storage calculation	At risk: 36 bytes fits EU868 but not US915 DR0 or AS923 DR2 (11 bytes); 22 months on 32 GB
R8	Fit without harming the structure	No drilling, welding or cutting; strain gauges only on an area of paint removed and recoated with the owner's permission; fitted by two trained people in 4 h or less from below or beside the deck	Fitting sequence estimate; later trial fit	Not verifiable at TRL 3 for time (2.8 h estimate); no drilling or welding met by design
R9	Keep clear of what passes under and over the bridge	Nothing more than 15 mm below the girder soffit; nothing on the deck side except the FieldNode on a handrail post, with tamper-resistant fasteners	Model check	Met on paper (10 mm; hub inside the girder outline)
R10	Survive outdoors	Hub IP67, FieldNode IP65; operate from -20 to $+50^\circ\text{C}$; 5 years with one gauge recoat and no cell replacement	Datasheets and design review; later environmental checks	At risk: FieldNode heat (FND-CAL-001) and gauge coating life
R11	Protect privacy	No cameras or microphones; only structural readings leave the device	Design review	Met by design
R12	Affordable		Priced BOM (bom/ bom.csv)	At risk: \$244.00 (2.4 % margin); complete \$370.00

ID	Requirement	Target	Verification	Status at TRL 3
		BridgePulse-specific parts \$250 or less, with the complete cost including the FieldNode core stated beside it		
R13	Present results responsibly	Every output labeled as monitoring data that supports inspection; no pass or fail safety rating anywhere in the system	Review of dashboard and report templates	Met by design intent; not yet implemented

■ Requirements not met or at risk

- **R2 (repeatability) not met:** on a light footbridge the mass of each walker lowers the first mode by up to 6.3 %, so the hourly estimate moves with traffic. Proposed responses are in `docs/REVIEW.md`, awaiting Amish.
- **R3 (change detection) not verifiable at TRL 3:** it depends on modeling temperature and load to a daily residual of 0.75 % or less.
- **R5, R7, R10 and R12 at risk:** cold-range temperature accuracy, US915 and AS923 payload limits, FieldNode heat and coating life, and a thin cost margin on indicative prices.
- **R4 and R8 not verifiable at TRL 3:** gauge drift and fitting time need tests or a trial fit.

■ Assumptions

- The accelerometer follows the ADXL355 datasheet summary ([Analog Devices](#)); a different part must meet R1.
- The FieldNode core offers 100 mW average for sensors as its design value (115 mW ceiling, FND-CAL-001) and two sealed M12 5-pin ports with a switched rail. BridgePulse assumes RS-485 and a 5 V rail on FieldNode's candidate pinout, which is still open (BRP-DDR-001, O1).
- A temperature-compensated frequency baseline needs at least four weeks of data before any flag, following the approach of [Peeters and De Roeck \(2001\)](#).
- Steel modulus 210 GPa for converting strain to stress. The other calculation assumptions are in BRP-CAL-001, Table 1.